

PRANAY KAPOOR

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EDUCATION

Manipal Institute of Technology

B.Tech in Computer Science (Honours)

- CGPA: 8.68 / 10 | Scholars Scholarship recipient (top 9% of cohort)

Bengaluru, India

Jul 2023 – Present

TECHNICAL SKILLS

Languages: Python, C, C++, Java

Core CS: Data Structures & Algorithms, OOP, Computer Architecture, Compiler Design

Systems: Processes, Threads, Memory Management, CUDA, MPI, Asyncio

Data & Tools: NumPy, Pandas, SQL, MongoDB, Git, Docker

ML / AI: PyTorch, TensorFlow, HuggingFace Transformers, Scikit-learn, OpenCV, MediaPipe

EXPERIENCE

AI/ML Intern

Jun 2025

BISAG-N, Ministry of Electronics & IT (MeitY), Government of India

Gandhinagar, India

- Built a full-stack medical imaging platform (React/Vite + Flask) for binary pneumonia detection and 4-class brain tumor classification (Glioma, Meningioma, Pituitary, Normal); MobileNet achieved 90.06% accuracy and AUC-ROC of 0.97 on a 624-image chest X-ray test set.
- Benchmarked 5 architectures (baseline CNN, CNN + class weights, CNN + Squeeze-and-Excitation, ResNet50, MobileNet) for pneumonia and 3 for brain tumor (CNN, MobileNetV2, EfficientNetB0); implemented an ensemble voting mechanism with MobileNet tie-breaker fallback.
- Integrated Grad-CAM explainability pipeline surfacing pathological heatmap overlays on X-rays and MRIs; added OOD detection to reject non-medical inputs and flag low-confidence predictions (<85%) for manual review.

PROJECTS

Apiro | Python, LLaMA 3.1 8B, HuggingFace, Shannon Entropy, Bioinformatics

2026

- Directed belief graph engine that uses LLM token-entropy as a traversal heuristic over clinical pathways and gene-disease target spaces; epistemic saturation acts as a self-terminating stopping condition, and entropy reversal flags circular reasoning — both validated against differential diagnosis literature.
- Ran calibration experiments to select LLaMA 3.1 8B (T=0.7) over Mistral 7B; ablation confirmed token-entropy as the sole necessary signal by eliminating semantic dispersion. Integrates MedRAG, HPO, ClinVar, and OpenFDA as evidence sources. *[Add any convergence or runtime numbers here if available]*

NexaCred | Python, LightGBM, FastAPI, Docker

2025

- Trained a LightGBM credit scoring model on class-imbalanced DeFi wallet histories; achieved F1 = 0.992 and G-mean = 0.918 with dynamic risk-adjusted interest rate computation exposed via a FastAPI service.
- Owned the ML pipeline end-to-end within a group project: REST API integration, Docker containerisation for reproducible deployment, and interface to the broader application stack.

Genetic Feature Selection & Hyperparameter Optimizer | Python, Scikit-learn, NumPy, Pandas | GitHub

2026

- Genetic algorithm framework for joint feature selection and hyperparameter tuning across KNN, SVM, and Random Forest; fitness function weighted accuracy at 99% and feature-reduction ratio at 1%.
- Achieved +9.1% accuracy on Ionosphere (KNN) and +1.6% on Breast Cancer (KNN & RF) while dropping up to 22 redundant features over 10 generations.

Face Emotion Recognition | PyTorch, ResNet-18, MediaPipe, Scikit-learn, OpenCV, Streamlit | GitHub

2026

- Two-pipeline classifier balancing latency against accuracy: Pipeline 1 extracts MediaPipe facial landmarks → geometric ratios → Random Forest (<5 ms CPU inference); Pipeline 2 runs a grayscale ResNet-18 fine-tuned on FER-2013 with class-weighted cross-entropy loss.
- Reconfigured ResNet-18 input from 3-channel RGB to 1-channel grayscale, reducing input tensor size by 66%; dynamic class weighting improved minority-class recall (Disgust, Fear) by 18%.

Async API Caching Service (Arrodes) | Python, Asyncio, NLTK, BeautifulSoup4, Docker | GitHub

2024

- Two-tier cache (in-memory + file-based) reduced single-request latency from 4.67 s to 76 ms (98% reduction), eliminating redundant network fetches and bypassing external rate limits.
- Offloaded blocking HTTP and parse tasks to asyncio thread pools; stress-tested under concurrent load — 9.5× higher throughput and 94% lower average latency versus uncached baseline.